6. Draw a right triangle whose hypotenuse is 10 cm and one of the legs is 8 cm. Locate its incentre and also draw the incircle.

7. Draw $\Delta ABC$ given $AB = 9$ cm, $\angle CAB = 115°$ and $\Delta ABC = 40°$. Locatge its incentre and also draw the incircle. (Note: You can check from the above examples that the incentre of any triangle is always in its interior).

8. Construct $\Delta ABC$ in which $AB = BC = 6$cm and $\angle B = 80°$. Locate its incentre and draw the incircle.

## Exercise 4.7

### Multiple Choice Questions

1. The exterior angle of a triangle is equal to the sum of two

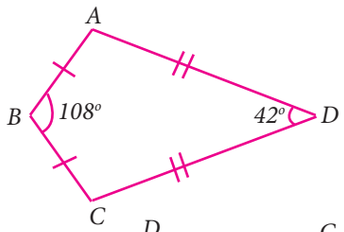
   (1) Exterior angles (2) Interior opposite angles

   (3) Alternate angles (4) Interior angles

2. In the quadrilateral $ABCD$, $AB = BC$ and $AD = DC$ Measure of $\angle BCD$ is

   (1) 150° (2) 30°

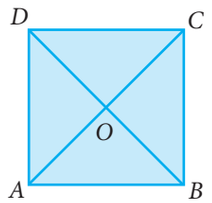
   (3) 105° (4) 72°




3. $ABCD$ is a square, diagonals $AC$ and $BD$ meet at $O$.
   The number of pairs of congruent triangles with vertex O are

   (1) 6 (2) 8

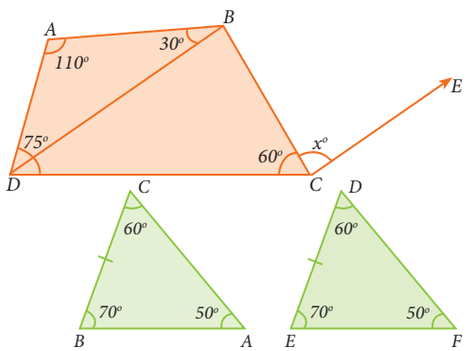
   (3) 4 (4) 12




4. In the given figure $CE \parallel DB$ then the value of $x^o$ is

   (1) 45° (2) 30°

   (3) 75° (4) 85°




5. The correct statement out of the following is

   (1) $\Delta ABC \cong \Delta DEF$ (2) $\Delta ABC \cong \Delta DEF$

   (3) $\Delta ABC \cong \Delta FDE$ (4) $\Delta ABC \cong \Delta FED$

6. If the diagonal of a rhombus are equal, then the rhombus is a

   (1) Parallelogram but not a rectangle

   (2) Rectangle but not a square

   (3) Square

   (4) Parallelogram but not a square



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